Classification Fine-Tuning Disrupts Differentiation Gradients in Single-Cell Foundation-Model Representations

Kunhui Xu[a,*], Qi Dai[a]

[a]*National Clinical Research Center for Ocular Diseases, Eye Hospital, Wenzhou Medical University, Wenzhou, Zhejiang 325035, PR China*

Abstract

Whether fine-tuning preserves the continuous differentiation geometry encoded in single-cell foundation models (scFMs) has not been quantified. Using the human corneal epithelium as an ideal test system whose four-layer differentiation continuum (limbal stem cells → transit-amplifying cells → basal → differentiated) has a known a priori ordinal structure; we show that classification fine-tuning of Geneformer-V2-104M collapses the differentiation gradient (Spearman ρ 0.82 → −0.008; sign-inconsistent across seeds) while preserving discrete layer topology (layer adjacency accuracy ≈0.97 across methods). Continuity-Aware Fine-Tuning (CAFT) was evaluated under the same protocol and was not promoted: a post hoc audit of the archived training code showed that its continuity term never executed (a dead hook in the released Transformers version), so the $\lambda = 0.5$ checkpoints were in fact trained on pure cross-entropy, identical in objective to the $\lambda = 0$ arm; the higher final-layer ρ of that group (+0.210 mean) reflects run-to-run nondeterminism (bfloat16 training, stochastic data preparation) rather than a causal effect of the CAFT objective. Differentiation geometry and classification are nonetheless separable: naive fine-tuning collapsed the gradient while retaining classification performance (macro-F1 0.6492). We term this topology-preserving ordinal-axis collapse (TP-OAC): the ordinal axis collapses while topology survives. To quantify this dissociation, we constructed a source-disjoint external benchmark (GSE153515; two donors) with the locked evaluation protocol as a validation resource.

Keywords: single-cell foundation models, classification fine-tuning, differentiation gradient, corneal epithelium, continuity-aware fine-tuning, external benchmark

1. Introduction

The corneal epithelium is maintained by a tightly regulated differentiation continuum from quiescent limbal stem cells (LSCs) at the limbal palisades of Vogt, through transit-amplifying cells (TACs), to committed basal epithelial cells, and ultimately to terminally differentiated squamous cells forming the protective ocular surface barrier [1]. This ordered progression is functionally required: disruption at any stage (LSC deficiency, aberrant amplification,

*Corresponding author

Email address: dq@mail.eye.ac.cn (Kunhui Xu)

or premature terminal differentiation) compromises corneal transparency and may lead to irreversible vision loss [2].

Single-cell RNA sequencing (scRNA-seq) has resolved this continuum at single-cell resolution, with marker genes for LSCs (e.g., *TP63*, *BIRC5*, *KRT15*), TACs (e.g., *KRT14*, *CDKN1A*), and differentiated populations (e.g., *KRT12*, *KRT3*, *TGM1*) identified across independent studies [3–5] and aggregated in multi-tissue atlases such as the human ocular surface atlas in CELLxGENE Discover [6]. Yet these efforts produce study-specific embeddings that are not directly comparable across datasets, and no unified reference model maps individual cells to a shared, ordered differentiation-state space; the closest approach, a corneal cell-state reference model with support-vector-machine classification (cPredictor) [7], provides a supervised classifier rather than an evaluation of foundation-model representations or their geometric structure.

Foundation models pretrained on massive single-cell corpora offer a potential solution. Geneformer, a 12-layer Transformer pretrained on approximately 104 million cells from 2,903 datasets [8, 9], and related architectures such as scGPT [10] and scBERT [11], learn contextual gene–gene representations that transfer across tissues. Systematic evaluations have tempered expectations: a unified benchmark of thirteen single-cell foundation models found classical PCA remains competitive in several settings, with pretrained embeddings advantageous mainly in low-label and transfer-learning scenarios [12]; biology-driven re-analyses questioned the pre-train-then-fine-tune paradigm [13]; and large-scale evaluations concluded that frozen, zero-shot usage is generally not recommended for downstream prediction [14]. Benchmark-driven evaluation has become an established genre in this journal, spanning single-cell foundation models [15] and clinical question answering [16], and representation-quality assessment is emerging as a distinct concern in embedding evaluation [17]. Whether general-purpose representations adequately capture the ordered structure of a specific differentiation hierarchy, such as the corneal continuum, remains open; conversely, task-specific fine-tuning may improve discriminative accuracy while inadvertently collapsing the continuous geometric relationships among adjacent differentiation states that are biologically essential.

The fine-tuning literature has focused on whether adaptation improves classification accuracy and whether it causes forgetting of learned cell-type knowledge; full fine-tuning induces catastrophic forgetting at the label level, motivating parameter-efficient remedies [18]. With respect to geometry, a zero-shot benchmark reported geometric collapse of foundation-model embeddings during cellular dynamics reconstruction, with HVG/PCA outperforming them for trajectory recovery [19], and EMP-FM showed that naive fine-tuning often fails to resolve intermediate cell states under a dual classification–pseudotime objective [20]. Layer-resolved dissections of single-cell foundation models have begun to probe where biological information resides within the network [21, 22], and fine-tuned representations are deployed in practice without quantifying whether adaptation preserves ordinal structure [23]; a related cautionary literature documents cases in which well-performing models fail in settings that matter [24, 25]. What remains absent is a quantified measurement, under controlled fine-tuning conditions, of whether supervised adaptation preserves or destroys the continuous differentiation-state gradient that orders a hierarchy, which we term the *continuity problem*. The corneal epithelium is an ideal test system: its four ordered stages have well-established markers, so the expected ordinal structure is known *a priori* and gradient preservation can be measured directly in embedding space. Continuity-Aware Fine-Tuning

(CAFT), introduced here, augments the classification objective with a regularization term encouraging adjacent differentiation layers to remain proximal in embedding space; whether it preserves ordered geometry without sacrificing classification performance is a hypothesis to be tested on externally held-out, source-disjoint data, not an assumption.

Here we quantify the effect of classification fine-tuning on differentiation-state geometry in the normal human corneal epithelium. Using a donor-disjoint fair-split protocol on a 10-study corneal and ocular surface atlas (99,606 training cells from 49 donors; 20,394 test cells from 25 donors; zero donor overlap), we compare four configurations sharing identical preprocessing, data splits, classification head, and training budget: (i) HVG/PCA with a linear classifier, (ii) frozen Geneformer-V2-104M with a linear probe, (iii) Geneformer with standard supervised fine-tuning ($\lambda = 0$), and (iv) Geneformer with CAFT ($\lambda = 0.5$), all evaluated on a fully independent, source-disjoint external dataset (GSE153515; 17,587 cells, 2 donors). Our contributions are threefold: (1) a quantified measurement, to our knowledge the first in the normal human corneal epithelium, of fine-tuning-induced differentiation-gradient collapse and its dissociation from preserved discrete layer topology, layer-resolved across all thirteen hidden states and localized to the Transformer layers nearest the classification head; (2) an honest negative result against a pre-registered four-criterion promotion rule: CAFT as implemented was not promoted, and a post hoc audit showed that its continuity term never executed, so the promotion test did not exercise the CAFT objective; the result is paired with a constructive layer-resolved diagnostic, in the spirit of recent critical-evaluation studies in this journal [25, 26]; (3) a source-disjoint scFM benchmark for the normal human corneal epithelium, comprising a training atlas, external test set, and locked multi-seed evaluation protocol, as a validation resource for future methods. These measurements define a collapse mode we term *topology-preserving ordinal-axis collapse* (TP-OAC), in which the ordinal differentiation axis is erased while discrete layer topology survives; it is distinct from the ordinal geometry that ordinal supervision induces [27], from pretrained layer profiles characterized without adaptation [28], and from the trajectory-fidelity trade-offs that trajectory benchmarks quantify across embedding methods [29]. Different from prior evaluations of embedding quality, which focused on visualization stability [17] or zero-shot geometry [19], our measurements quantify how supervised adaptation itself alters ordinal differentiation geometry under controlled fine-tuning conditions. Our scope is deliberately restricted to normal human corneal epithelium; evaluation of disease states, including dry eye disease, is deferred to future work.

2. Results

2.1. Fine-tuning collapses the differentiation gradient while preserving layer topology

We first asked whether task-specific fine-tuning of a single-cell foundation model preserves the ordered geometry of the corneal differentiation continuum. Four model arms sharing identical preprocessing, data splits, classification head, and training budget (HVG/PCA with a linear classifier; frozen Geneformer-V2-104M with a linear probe; naive fine-tuning with $\lambda = 0$; and Continuity-Aware Fine-Tuning, CAFT, with $\lambda = 0.5$, the latter two trained with three random seeds 42, 123, and 456) were evaluated on a source-disjoint external test set (GSE153515; Figure 1).

The two non-adapted reference arms preserved strong differentiation-gradient rank correlation on external data: Spearman ρ was 0.8206 for HVG/PCA and 0.8064 for frozen Geneformer (Table 1). By contrast, naive supervised fine-tuning collapsed this ordinal signal to near zero: per-seed ρ was -0.0424 (seed 42), -0.0360 (seed 123), and $+0.0539$ (seed 456), with a three-seed mean of -0.0082. The sign of ρ was inconsistent across seeds, indicating that the residual ordinal correlation after fine-tuning was indistinguishable from noise; bootstrap 95% confidence intervals for the collapsed values all spanned zero (Figure 5, Table 1).

Despite this gradient collapse, discrete layer topology was preserved. Layer adjacency accuracy (LAA) remained high in all four arms: 0.9771 (HVG/PCA), 0.9665 (frozen Geneformer), 0.9768 (naive, three-seed mean), and 0.9773 (CAFT, three-seed mean). Classification fine-tuning therefore disrupted the continuous ordering of the differentiation continuum while leaving local layer-adjacency structure essentially intact: continuous gradient geometry and discrete layer topology are dissociable under task adaptation.

The $\lambda = 0.5$ arm must, however, be read as a failed control: an audit of the six archived training scripts found them byte-identical to the current scripts/caft_gf.py and showed that the continuity term never executed in any run (Methods, Provenance and contamination audit). In the Transformers release used (4.57), the "on_epoch_begin" hook fires only for registered Trainer callbacks; CAFTTrainer.on_epoch_begin is a plain method that no callback ever invokes, layer centroids were never populated, and the logged training history of every run contains only the plain cross-entropy "loss" key. The $\lambda = 0.5$ checkpoints therefore trained pure cross-entropy, identical in objective to the $\lambda = 0$ arm. The separation between the two groups' final-layer ρ (naive -0.0424, -0.0360, $+0.0539$; mean -0.0082; $\lambda = 0.5$ $+0.1913$, $+0.2298$, $+0.2095$; mean $+0.2102$) is attributable to run-to-run nondeterminism, bfloat16 training over 7,473 steps and stochastic data preparation that permutes class identifiers and row order between runs, not to a causal effect of λ. With three runs per group, the complete 3-versus-3 separation carries no statistical interpretation (two-sided Mann-Whitney $p = 0.1$), and no claim of a continuity-regularization effect is supportable. The classification comparison stands: macro-F1 was 0.6451 (95% CI 0.6164–0.6738) for the $\lambda = 0.5$ group versus 0.6492 (95% CI 0.6233–0.6751) for naive fine-tuning ($\Delta = -0.0041$; overlapping intervals), with no effect of λ. Differentiation geometry and classification performance are nonetheless separable in the direction established by the naive arm: fine-tuning achieved the highest classification accuracy while collapsing the gradient to noise.

2.1.1. Layer-resolved analysis localizes the collapse to upper transformer layers

To determine whether fine-tuning erases the differentiation-gradient signal globally or damages it in a layer-localized manner, we re-measured ρ at every hidden state of the Transformer, thirteen in total (embedding output L0 and layers L1–L12), for all seven Geneformer-based configurations on the 14,775 external-test epithelial cells (Methods). A sanity gate required each model's L12 ρ to reproduce the corresponding single-value measurement within 10^{-6}: all seven models passed bit-exactly. At L0 the position-0 [CLS] representation is constant across cells, so ρ is undefined (NaN) and was excluded (expected behavior).

Three findings emerged (Figure 5). First, early-layer representations preserved the gradient in every arm: L1–L4 ρ lay between 0.41 and 0.84, and the fine-tuned models reached their maximum at L3 (0.8380, 0.8062, 0.8382 for naive seeds; 0.8339, 0.8100, 0.8368 for CAFT seeds), matching or exceeding the frozen model at the same depth (L3, 0.7656). Second,

the collapse observed at the final layer of the naive models was concentrated in the upper half of the network: from L5 (0.3973, seed 42) ρ decayed toward zero or negative values at L12, whereas the frozen model's gradient signal rose to its network-wide maximum of 0.8064 precisely at L12. Third, the $\lambda = 0.5$ checkpoints ended at L12 ρ of 0.1913–0.2298, above the naive group's near-zero values but far below the frozen model's 0.8064; this difference is not attributable to the CAFT objective, which never ran (Methods, Provenance and contamination audit).

Bootstrap confidence intervals at L12 (n = 1,000, cell-level resampling) quantified the per-checkpoint values: frozen $\rho = 0.8064$ [0.7918, 0.8164]; naive seeds all spanned zero (-0.0424 [-0.1070, 0.0318]; -0.0360 [-0.1074, 0.0406]; $+0.0539$ [-0.0523, 0.1293]); the $\lambda = 0.5$ checkpoints were positive (0.1913 [0.1597, 0.2211]; 0.2298 [0.1702, 0.2760]; 0.2095 [0.1708, 0.2419]) and, in this sample, separated from both groups. Because the continuity term never executed in this arm (Methods, Provenance and contamination audit), that separation is a property of these particular cross-entropy runs, not evidence of a CAFT effect. Gradient collapse under classification fine-tuning is therefore a head-proximal overwrite: the signal persists in intermediate layers and is selectively disrupted in the layers nearest the classification head.

To test whether this overwrite reflects the training process rather than an artifact of the trained classification head, we performed a randomized-head ablation on the naive seed-42 checkpoint: replacing the trained head with a fresh random head (no training) left the layerwise differentiation-gradient ρ bit-identical to the intact model at all thirteen layers (L12 $\rho = -0.0424$; maximum hidden-state difference 0), while the same random-head treatment applied to the frozen body preserved the frozen profile exactly (L12 $\rho = 0.8064$). The L5–L12 collapse is therefore intrinsic to the fine-tuned body weights, a training-induced rewrite of head-proximal layers, and not a readout artifact of the trained head.

Trajectory inference confirmed the geometric dissociation at the level of inferred cell ordering (Figure 6). Diffusion pseudotime on the same external cells, rooted at the LSC class, correlated strongly with the reference layer-ordering pseudotime for the frozen model ($\rho = 0.850$, 95% CI 0.842–0.858) but weakly for naive fine-tuning ($\rho = 0.323$–0.348), with CAFT intermediate ($\rho = 0.372$–0.428). The loss of ordinal information under naive fine-tuning is thus observable not only in our gradient metric but in a widely used trajectory-reconstruction procedure.

2.2. A source-disjoint external benchmark as validation resource

2.2.1. Dataset construction and external test annotation

The training atlas was assembled from ten constituent studies spanning limbus, cornea, sclera, and fetal ocular surface tissue (CELLxGENE Discover; Figure 1), yielding 120,000 cells with curated cell-type labels. A donor-disjoint fair-split protocol (seed = 42) allocated donors deterministically to training (99,606 cells from 49 donors) and internal test (20,394 cells from 25 donors) partitions, with zero donor overlap and a SHA-256-locked manifest; after quality filtering, 79,684 training cells were used for model fitting.

The external test set (GSE153515) comprised 17,587 cells from two donors profiled by 10x Genomics from normal human limbal basal epithelium, confirmed source-disjoint from the training collection. Following quality control (15,827 retained cells; 90.0%) and unsupervised integration and clustering, clusters were annotated by differential-expression enrichment and

harmonized to four ordered classes: basal (6,698 cells), differentiated (4,483), LSC (3,423), and TAC (171); 1,052 non-epithelial cells were excluded (Figure 2). All annotation parameters were locked before evaluation.

2.2.2. Fine-tuning drives external performance; the internal baseline ranking inverts

On the external test set, the HVG/PCA reference arm achieved donor-level macro-F1 of 0.6192 (95% CI 0.5747–0.6637), outperforming frozen Geneformer (0.4421; 95% CI 0.3749–0.5093), the opposite of the internal training-split ranking (0.8834 versus 0.7787). Frozen Geneformer showed poor calibration on external data (ECE 0.3010) and basal-class collapse (recall 0.216), whereas HVG/PCA remained well calibrated (ECE 0.0117) with strong gradient correlation. Both fine-tuned arms (naive 0.6492; CAFT 0.6451) outperformed both baselines on external data. These results suggest that fine-tuning, not pretraining alone, drives cross-dataset generalization (Figure 3). The magnitude of these external scores sits in the range expected for donor-disjoint validation: cross-dataset macro-F1 values fall well below internal-split numbers across the single-cell foundation-model literature, which consistently reports macro-F1 "much lower" than accuracy once evaluation leaves the training cohort [13, 30].

2.2.3. Donor-level consistency and robustness to the rare TAC class

Both adapted arms generalized across the two external donors (donor_1 > donor_2 in all six runs; Table 1). Because the TAC class is extremely rare (171/14,775 cells; 1.2%) and received zero true-positive predictions in every adapted run, we performed a sensitivity analysis excluding TAC (Figure 7). Three-class macro-F1 rose by +0.21–0.22 in all adapted arms (naive 0.8599–0.8763; CAFT 0.8479–0.8786) and by +0.15 for frozen Geneformer (0.5921); the three-class confidence intervals did not overlap the four-class intervals, confirming that TAC rarity systematically depresses the primary endpoint. The naive-versus-CAFT comparison remained non-significant under the three-class definition, and the arm ranking was unchanged.

2.3. CAFT fails pre-registered promotion criteria: an honest negative

The pre-registered CAFT promotion rule required all four criteria to be satisfied simultaneously (Methods). Because the archived $\lambda = 0.5$ training never executed the continuity term (Methods, Provenance and contamination audit), the promotion comparison did not actually test CAFT: Criteria 1 and 3 (external continuity improvement and directional consistency) compare two pure-cross-entropy groups, and the +0.218 ρ separation with positive values in all three $\lambda = 0.5$ seeds reflects run-to-run nondeterminism rather than the CAFT objective. Criterion 2 (macro-F1 non-inferiority) was satisfied: the mean decrease of 0.004 fell within overlapping bootstrap intervals. Criterion 4 (provenance audit) passed with limitation: the V1 (30M) corpus contained zero corneal datasets (0/8 GEO accessions matched), but the V2 (104M) audit was not executable because no public machine-readable manifest exists. Accordingly, CAFT as implemented was not promoted and provides no evidence that a continuity-regularized objective preserves gradient geometry; the test of a correctly wired continuity objective was carried out in the ordinal arms below, and the layer-aware HPA-FT arms remain pending.

2.4. An ordinal head parameterization preserves the differentiation gradient

The honest negative leaves open whether any fine-tuning recipe can preserve the differentiation gradient at matched accuracy. We therefore tested a constructive alternative:

reparameterize the classification head along the known ordinal axis instead of treating the four layers as unordered classes. Following the cumulative ordinal regression (CORN) formulation [31], the head emits $K - 1 = 3$ binary threshold logits, where threshold k predicts whether the cell has reached ordinal stage k or beyond; the loss is the sum of the three binary cross-entropies (masked to epithelial cells), and inference recovers the predicted layer as the number of thresholds with sigmoid activation ≥ 0.5. The CORN head makes no equidistance assumption on the thresholds and yields class probabilities directly for calibration ($P(y = k) = p_k - p_{k+1}$).

Two ordinal arms were trained under the identical locked protocol and budget: ordinal-naive ($\lambda = 0$; ordinal loss only) and ordinal-CAFT ($\lambda = 0.5$; ordinal loss plus the CAFT continuity term, here wired through a registered Trainer callback and therefore genuinely executed, in contrast to the archived CAFT arm). Both preserved the differentiation gradient at classification accuracy matched to the softmax arms (Table 1): three-seed mean ρ was 0.8330 (per-seed 0.8299–0.8351) for ordinal-naive and 0.8277 (0.8242–0.8321) for ordinal-CAFT, versus -0.008 for naive softmax fine-tuning, while donor-level macro-F1 was 0.6490 (95% CI 0.6100–0.6879) and 0.6479 (95% CI 0.6099–0.6860), statistically indistinguishable from the naive arm's 0.6492. Layer adjacency accuracy remained $\approx$0.97 in both ordinal arms.

The near-equality of the two ordinal arms ($\Delta\rho = 0.005$) attributes the gradient preservation to the ordinal head parameterization itself rather than to the continuity term: encoding the a priori layer order as ordered thresholds is sufficient, and the correctly wired centroid-adjacency regularization adds no measurable benefit on this axis. This is the constructive counterpart to the CAFT honest negative: a head reparameterization that respects the known ordinal structure preserves gradient geometry at matched classification performance, whereas the unordered softmax head erases it (Figure 4).

Table 1: External test performance by model arm (GSE153515, four classes). Adapted-arm values are three-seed means with donor-level bootstrap 95% CIs; reference arms are single runs (seed 42).

Model arm	Macro-F1	95% CI	LAA	ρ (ECE)
HVG/PCA + linear	0.6192	[0.5747, 0.6637]	0.9771	0.8206 (0.0117)
Frozen Geneformer	0.4421	[0.3749, 0.5093]	0.9665	0.8064 (0.3010)
Naive FT ($\lambda = 0$)	0.6492	[0.6233, 0.6751]	0.9768	-0.0082 (0.0575)
CAFT ($\lambda = 0.5$)	0.6451	[0.6164, 0.6738]	0.9773	+0.2102 (0.0600)
Ordinal-naive ($\lambda = 0$)	0.6490	[0.6100, 0.6879]	0.9739	+0.8330 (0.1251)
Ordinal-CAFT ($\lambda = 0.5$)	0.6479	[0.6099, 0.6860]	0.9749	+0.8277 (0.0609)

3. Discussion

The central finding of this study is that differentiation geometry and classification performance are separable properties of single-cell representations. Classification fine-tuning of Geneformer-V2-104M collapsed the differentiation-gradient Spearman ρ from 0.82 (0.8206 for HVG/PCA; 0.8064 for frozen Geneformer) to -0.008 (sign-inconsistent across seeds), while layer adjacency accuracy remained $\approx$0.97 in all adapted arms and macro-F1 rose to 0.6492. The $\lambda = 0.5$ arm, reported as CAFT, provides no evidence about the continuity objective: a

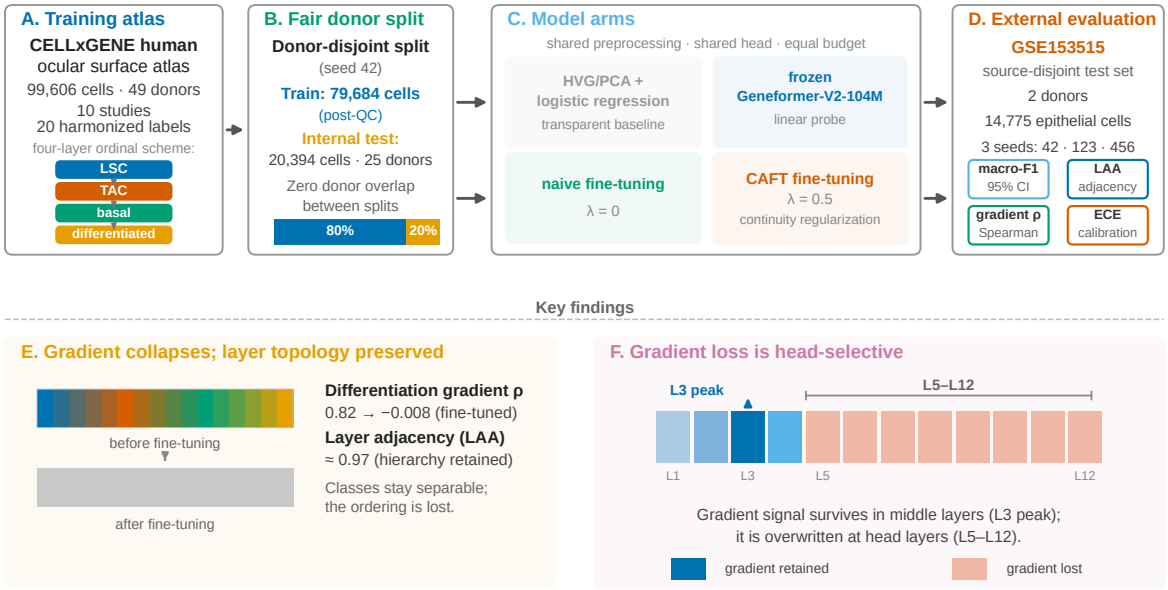


Figure 1: Study pipeline: donor-disjoint training atlas (CELLxGENE), locked evaluation protocol, four softmax-head model arms plus two ordinal-head arms, and source-disjoint external test (GSE153515).

post hoc audit of the archived training code showed that its continuity term never executed, so that arm trained pure cross-entropy, and the higher final-layer ρ of its checkpoints (+0.210 mean) is a run-to-run effect rather than evidence of restoration (Methods, Provenance and contamination audit). The layer-resolved analysis points to a head-proximal overwrite: the gradient signal persists in intermediate layers of the fine-tuned models, peaking at L3 with ρ of 0.81–0.84 and matching or exceeding the frozen model at the same depth (0.7656), is selectively degraded in the layers nearest the classification head (L5–L12). The frozen model, by contrast, reaches its maximum (0.8064) precisely at the final layer, indicating that pretraining concentrates differentiation structure in the readout layer and that classification training rewrites exactly that layer. This head-proximal overwrite account rests on the naive arm alone and is unaffected by the failed continuity control: the continuity objective never ran, so no claim about it can be drawn from these checkpoints, and the absence of classification benefit is a $\lambda = 0.5$ versus $\lambda = 0$ comparison of identical objectives. The practical implication for single-cell annotation practice is direct: pipelines that fine-tune foundation models for classification and then reuse the resulting embeddings for analyses that require continuous structure, such as trajectory inference, pseudotime ordering, or state-transition quantification, will systematically lose ordinal information even when classification accuracy is high, as our diffusion-pseudotime measurement confirms. Where continuous structure matters, intermediate-layer readouts, or frozen representations, deserve explicit consideration.

We frame this pattern as *topology-preserving ordinal-axis collapse* (TP-OAC): supervised adaptation erases the ordinal axis of a pre-existing differentiation continuum, cluster membership and layer topology survive, with the loss graded by depth and concentrated in head-proximal layers. Three comparisons locate the term. Ordinal supervision can induce ordered geometry, as in ordinal neural collapse (ONC) in cumulative-link regression, where

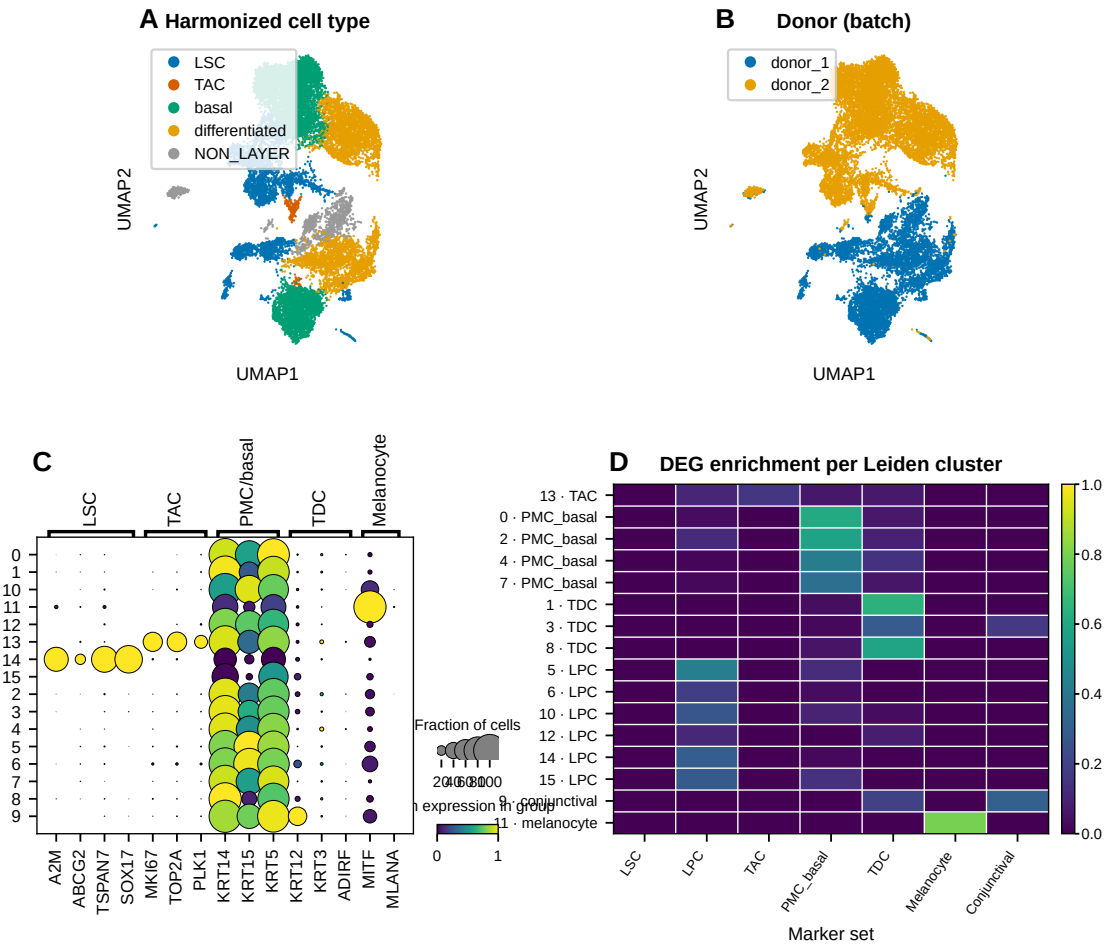


Figure 2: External test annotation (GSE153515): unsupervised integration, clustering, and DEG-based assignment to four ordered differentiation classes.

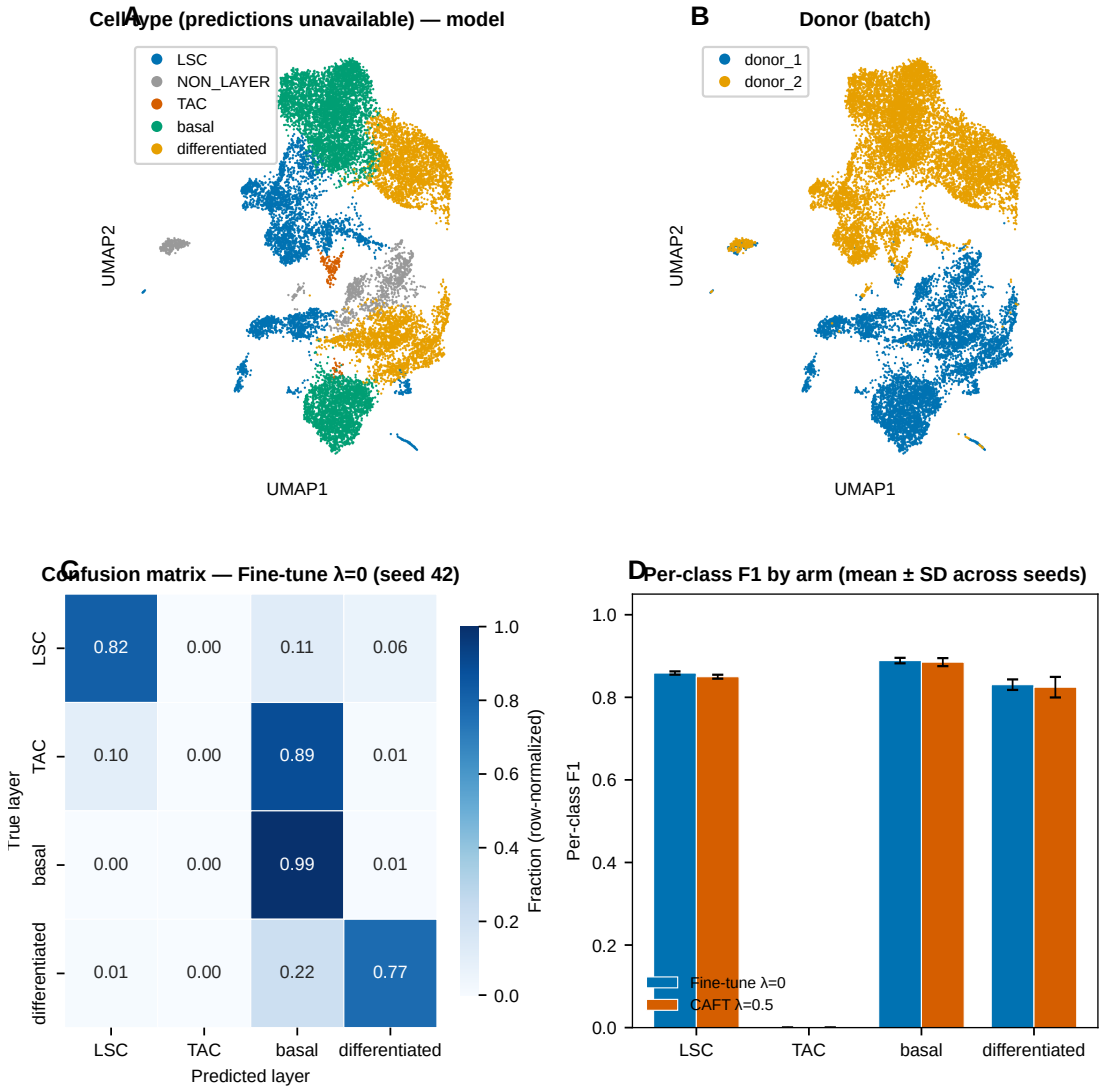


Figure 3: External evaluation of adapted models on GSE153515: prediction UMAP, donor UMAP, confusion matrix, and per-class F1 across seeds.

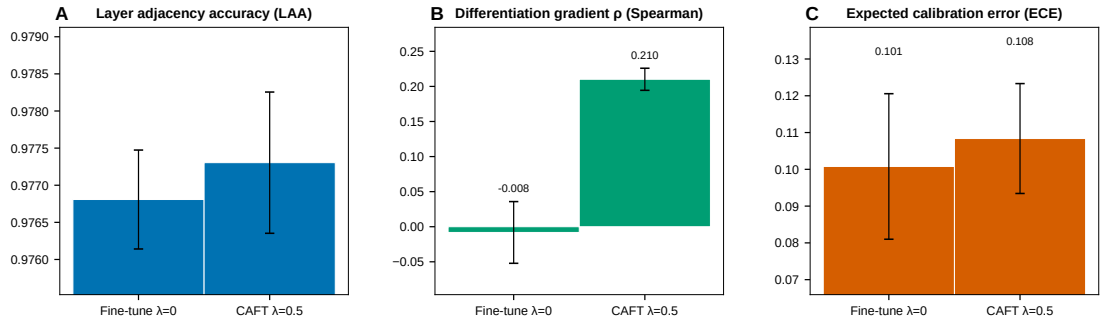


Figure 4: Continuity metrics under task adaptation: LAA, differentiation-gradient ρ, and ECE per seed for naive fine-tuning and CAFT.

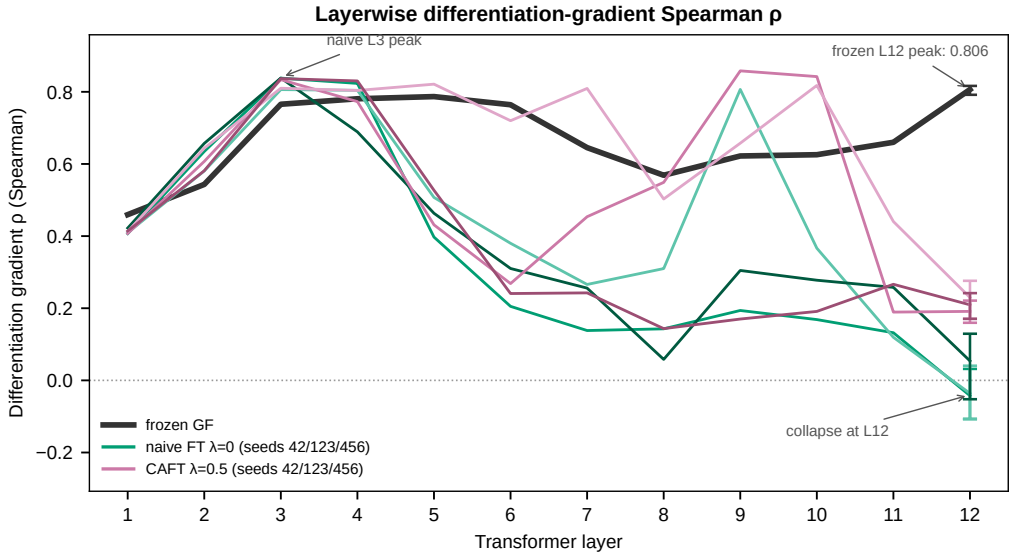


Figure 5: Layer-resolved differentiation-gradient Spearman ρ across thirteen hidden states for seven Geneformer-based configurations, with bootstrap 95% CIs at L12. The collapse is head-proximal: early layers retain the gradient, while L5–L12 decay.

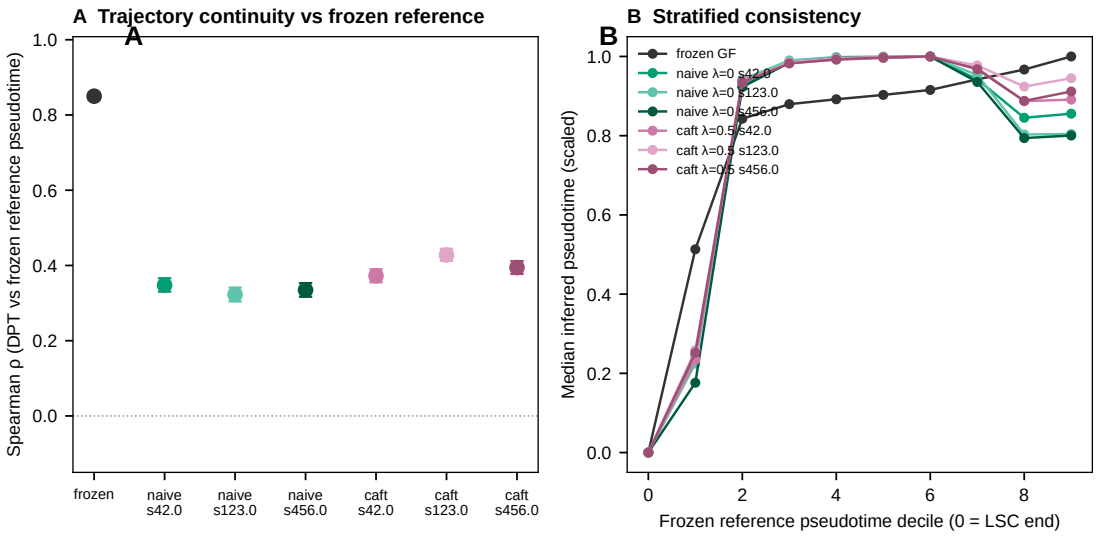


Figure 6: Trajectory inference (diffusion pseudotime) on external cells confirms the geometric dissociation: frozen representations preserve cell ordering relative to the reference pseudotime ($\rho = 0.850$), naive fine-tuning does not ($\rho \approx 0.33$), and CAFT is intermediate ($\rho \approx 0.40$).

class means align onto a one-dimensional axis with ordered thresholds, an ordering that holds only under fixed, equally spaced thresholds [27]; TP-OAC is the opposite failure mode, destruction of an existing ordinal axis by classification supervision, and our randomized-head ablation localizes it to the fine-tuned body. The acronym ONC also denotes an ordinal-collapse representation prior in visual navigation [32]; the phenomena are unrelated. Civale et al. characterized where differentiation signal resides in pretrained layer profiles under zero-shot conditions [28]; TP-OAC describes the change induced by fine-tuning, measured on the difference between pretrained and fine-tuned profiles rather than absolute profiles. The representational collapse R3F mitigates during fine-tuning is a generic degradation of pretrained features [33]; TP-OAC is ordinal-axis-specific and leaves discrete structure intact. To our knowledge, fine-tuning-induced collapse of an existing ordinal axis, with topology survival, has not previously been measured.

These findings connect to, and extend, several parallel lines of work. At the level of representation geometry, our layer-resolved ρ collapse is related to, but distinct from, the neural-collapse family: neural collapse is the terminal-phase convergence of within-class features observed during classification training [34], and related phenomena include the analogous collapse characterized in neural multivariate regression [35] and the feature distortion that fine-tuning can induce in pretrained representations [36]. Unlike these within-class convergence phenomena, TP-OAC is a cross-class ordinal-axis erasure that leaves discrete layer topology intact. That full fine-tuning preferentially rewrites head-proximal layers has long been documented in NLP [37]; our contribution is its first quantitative, layer-resolved measurement on an ordinal differentiation axis in a single-cell foundation model. Layer-selective freezing, as in surgical fine-tuning, targets out-of-distribution robustness rather than geometric fidelity [38]. Our measurements show these effects arising in a tissue-differentiation geometry, while discrete layer topology survives. Civale et al. showed that,

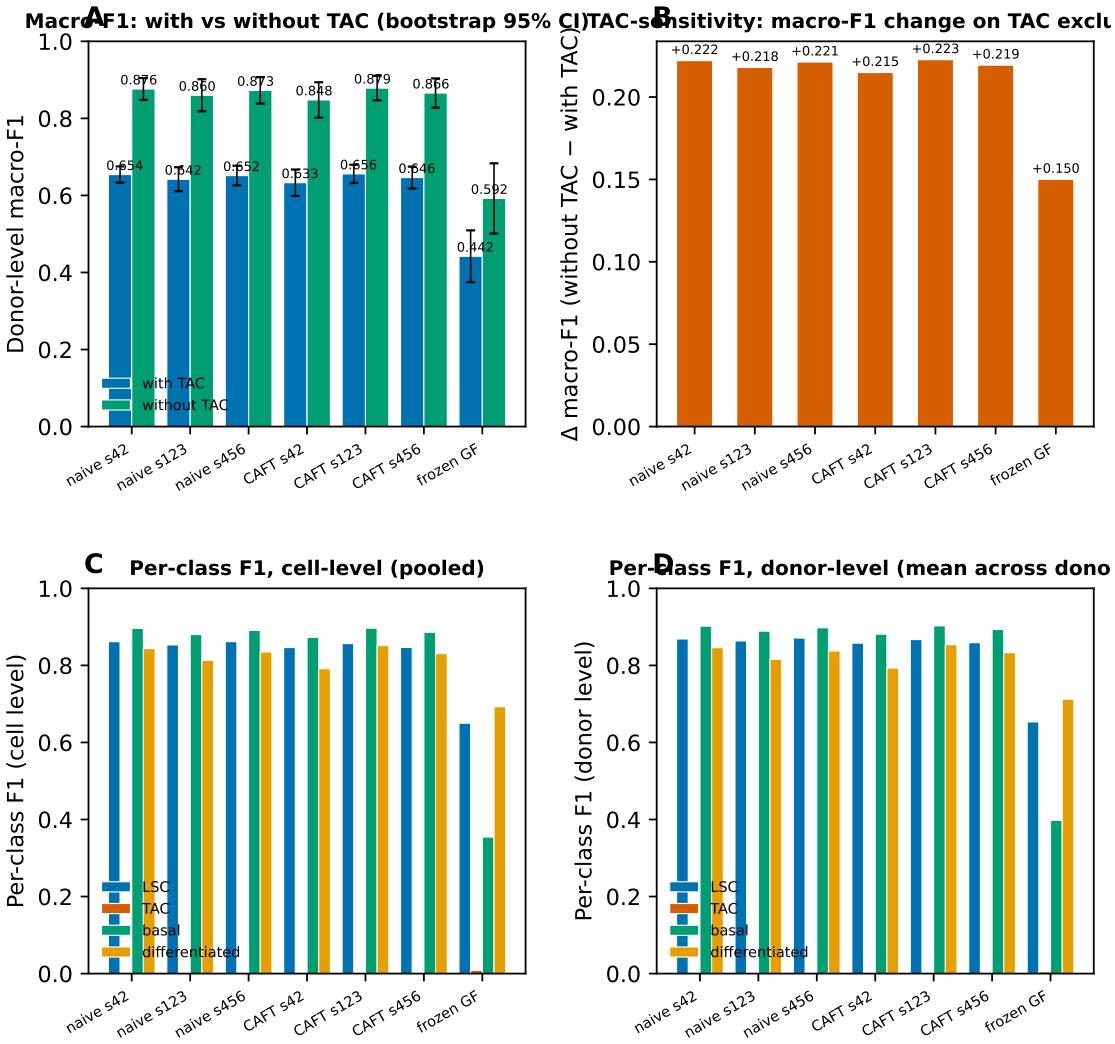


Figure 7: TAC sensitivity analysis: donor-level macro-F1 with and without the rare TAC class. Three-class confidence intervals do not overlap four-class intervals, confirming that TAC rarity depresses the primary endpoint; arm ranking is unchanged.

under zero-shot conditions, differentiation-related signal in single-cell foundation models concentrates at intermediate network depths rather than the final layer [28]; we add the orthogonal, fine-tuning dimension, showing that task adaptation selectively overwrites exactly the readout layers while intermediate layers retain the signal; the two observations together delineate where differentiation information resides and how adaptation destroys it. The original Geneformer analysis likewise inspected layer-wise representations and reported that successive layers progressively separate cell types [8]; that observation concerns discrete cell-type separation, whereas TP-OAC measures the fate of the continuous ordinal axis, which collapses rather than separates. Layer-resolved dissections of the pretrained models themselves report that biological information is organized across attention layers, that residual-stream geometry is fragile under data partitioning [21, 22], and that biological knowledge exhibits multi-dimensional spectral structure [39]; our measurements are complementary in that they quantify the *consequence of task adaptation* on ordinal geometry rather than the internal organization of the pretrained network. Whether pretrained encoders implicitly encode continuous progression from static observations has been answered affirmatively in computational pathology, where the diffusion-pseudotime trajectory fidelity of pathology foundation models predicts downstream classification [40]; our measurements address the analogous question for single-cell differentiation hierarchies and show that the pretrained gradient, though present, is not robust to classification adaptation. scPEFT demonstrated that full fine-tuning induces catastrophic forgetting at the label level [18]; our measurements show a deeper, geometric level of forgetting in which ordinal structure is erased even though classification performance improves, and even though discrete layer topology (LAA $\approx$0.97) survives. Objective-driven adaptation can also deliberately reorient latent geometry, as when a biological foundation model fine-tuned on perturbation profiles develops perturbation-centric axes [41]; our results document the complementary risk, in which classification supervision silently erases an ordinal axis. EMP-FM reported a positive result for a dual-task objective combining classification with pseudotime-guided regression [20], whereas our CAFT is an honest negative; a plausible explanation lies in the supervisory signal: pseudotime regression provides a direct ordinal target for every cell, whereas CAFT's centroid-based adjacency regularization constrains only the final representation, which, as our layer-resolved data show, is precisely the site already overwritten by the classification task. Finally, our external ranking (fine-tuned > HVG/PCA > frozen) aligns with the emerging consensus from field-wide benchmarking efforts [12–15] that frozen usage is generally inadequate and that fine-tuning benefits are setting-dependent; we accordingly do not claim foundation-model superiority over classical baselines, particularly given the internal-versus-external ranking inversion observed for the frozen arm (internal 0.8834 $\rightarrow$ external 0.4421).

Taken together, these results argue that classification accuracy alone is an insufficient criterion for evaluating adapted foundation models in differentiation biology: geometry-aware metrics, and layer-resolved readout in particular, expose losses that aggregate accuracy conceals. The source-disjoint benchmark released with this study (training atlas, external test set, and locked multi-seed evaluation protocol) is intended as a validation resource for such geometry-aware evaluation of future methods, in corneal biology and, we hope, beyond. Reporting the CAFT outcome as an honest negative, rather than a promoted safeguard, is deliberate: the pre-registered promotion rule was applied without post hoc tuning, and publishing such outcomes, paired with a constructive layer-resolved diagnostic, serves the

community better than selective reporting [25, 26].

3.1. Limitations of the study

This study has three principal limitations. First, it evaluates a single model family (Geneformer-V2-104M) on a single tissue (normal human corneal epithelium), and the external test comprises only two donors; whether the dissociation between gradient geometry and layer topology, and the head-proximal overwrite pattern, generalize across architectures, tissues, and larger donor panels remains to be tested. Second, the TAC class achieved F1 of 0.0 in every adapted run, reflecting extreme class rarity (171 of 14,775 external-test cells; 1.2%) rather than a method-specific failure; although our sensitivity analysis confirms that excluding TAC raises macro-F1 uniformly across arms without changing the ranking, conclusions regarding TAC representations are constrained by sample size. Third, the Genecorpus-104M (V2) pretraining contamination audit was not executable because no public machine-readable dataset manifest is available; the V1 audit (zero corneal datasets; 0/8 GEO accessions matched) provides only provisional reassurance, and pretraining contamination cannot be definitively excluded.

4. Methods

4.1. Data sources and preprocessing

Training atlas. Training and internal-test data were derived from the "Single cell atlas of the human ocular surface" collection (CELLxGENE Discover, collection identifier 0f7d022a-46c7-4e64-be4c-e34adbb78089). The data file used is the current dataset-version of CELLxGENE dataset 8ef06ea4-8d0f-4303-a652-7d4604402b1f (schema 7.1.0, published 2025-05-05), distributed as file 30127f02-d0b9-4b56-bf4c-e5b77e4989bd.h5ad (10.29 GB; MD5 39f098367af95bc3e1ee92528ceb251e; the file identifier is the dataset-version identifier at time of download, retained for traceability). The atlas is a pre-publication consortium product (CZI Cell Science / Human Cell Atlas Eye Network) without an associated journal publication or DOI, and is therefore cited as a CELLxGENE collection [6]. The collection aggregated ten constituent studies spanning corneal, limbal, scleral, and fetal ocular surface tissues [3–5, 42–45]. The aggregated AnnData object contained 120,000 cells with 20 curated cell-type labels and 31 author-assigned cell-type labels.

Cell-type labels were harmonized into a five-layer ordinal scheme reflecting the corneal epithelial differentiation continuum: LSC (including LPC and fetal epithelial progenitors; layer index 0), TAC (limbal epithelium; index 1), basal (Epi-PMC; index 2), wing (Epi-TDC/PMC; index 3), and squamous (Epi-TDC/unknown epithelium; index 4). Non-epithelial populations (immune, endothelial, stromal, conjunctival, melanocyte) were assigned layer index -1 (NON_LAYER) and excluded from ordinal analyses. For the four-class classification task evaluated on external data, the scheme was collapsed to LSC, TAC, basal, and differentiated (wing and squamous merged).

External test dataset. The external test set comprised GSE153515 (GEO accession), a 10x Genomics scRNA-seq dataset of normal human limbal basal epithelium [3, 4]. The dataset contained 17,587 cells from two donors (GSM4646295, GSM4646296) with an initial gene matrix of 32,738 features. Source disjointness from the training collection was verified programmatically via the CELLxGENE Discover API: no overlapping GEO accession, donor

identifier, or dataset hash was found between GSE153515 and any of the ten constituent training studies.

4.2. Fair donor-disjoint splitting protocol

To prevent donor-level information leakage between training and evaluation, a strict donor-disjoint split was applied to the training atlas prior to any model fitting. Donors were partitioned using a deterministic random assignment (seed = 42), yielding 49 donors (99,606 cells) for training and 25 donors (20,394 cells) for internal testing, with zero donor overlap between partitions. The split protocol ensured representation of all 20 curated labels and all 31 author-assigned labels in both partitions. The split manifest, including per-partition cell counts, donor lists, label distributions, and SHA-256 checksums of the resulting train and test AnnData files, was locked and archived (artifacts/paper1_fair_split/manifest.json). No cells were reassigned after manifest generation.

4.3. External test annotation pipeline

Because GSE153515 was deposited with an empty observation metadata table, a fully reproducible de novo annotation pipeline was implemented (scripts/protocol_15_annotate_gse153515.py): (1) donor assignment from the 10x barcode suffix convention; (2) marker scoring on the full 32,738-gene matrix using eight curated marker gene sets spanning LSC, TAC, basal, differentiated, conjunctival, melanocyte, immune, and stromal identities; (3) quality-control filtering by minimum gene count and maximum mitochondrial fraction (15,827 of 17,587 cells retained, 89.9%, spanning 10,605 genes); (4) integration and clustering—highly variable gene selection, log-normalization, scaling, PCA, Harmony batch correction (n_pcs = 30) [46], and Leiden clustering (resolution 0.6) yielding 16 clusters; (5) DEG-based cell-type assignment via Wilcoxon rank-sum test with an enrichment score $(n_{\text{in top200}}/n_{\text{total}}) \times (1 - \text{mean_rank}/200)$ against the top 200 cluster-specific DEGs, with clusters below an epithelial enrichment threshold of 0.3 classified as NON_LAYER; and (6) harmonization to the four-class scheme (basal 6,698; differentiated 4,483; LSC 3,423; TAC 171 cells) with non-epithelial cells excluded. All annotation parameters were locked in a configuration file (config/external_test_gse153515.yml) before any model evaluation.

4.4. Model architectures

Four model configurations were evaluated under an identical experimental protocol (same preprocessing, data split, classification head, training epochs, and random seeds); two additional ordinal-head arms (Arms 5–6) shared the same protocol and budget:

Arm 1—HVG/PCA + linear classifier (transparent baseline). Highly variable genes were selected from the training partition; expression values were log-normalized and scaled, PCA was applied, and a linear classifier was trained on the resulting principal components.

Arm 2—Frozen Geneformer + linear probe. Geneformer-V2-104M (BertForMaskedLM architecture; hidden dimension 768, 12 layers, 12 attention heads, vocabulary 20,275) was loaded from the Hugging Face model hub (snapshot commit 04c2b2e84da7c0f385c3f9ad8f3ec24bab6650e5; model weights SHA-256 fff5cba2...), pretrained on Genecorpus-104M (∼104 million cells from 2,903 datasets) [8, 9]. All Transformer

parameters were frozen; only a linear classification head appended to the [CLS] token representation was trained.

Arm 3—Geneformer naive fine-tuning ($\lambda = 0$). The full model was fine-tuned end-to-end with a standard cross-entropy classification loss (no continuity regularization), serving as the fair adaptation comparator.

Arm 4—Geneformer + CAFT ($\lambda = 0.5$). The full model was fine-tuned with the CAFT composite loss (Section 2.5) at a single fixed regularization strength, testing CAFT as a candidate safeguard for ordered differentiation geometry.

Arm 5—Geneformer + ordinal-naive head ($\lambda = 0$). The full model was fine-tuned end-to-end with the CORN ordinal loss (Section 2.5) and no continuity regularization.

Arm 6—Geneformer + ordinal-CAFT head ($\lambda = 0.5$). The full model was fine-tuned with the CORN ordinal loss plus the CAFT continuity term, wired through a registered Trainer callback so that the continuity objective was genuinely executed (in contrast to the archived CAFT arm).

Arms 3–6 were each trained with three independent random seeds (42, 123, 456) for three epochs. Training used a per-device batch size of 8 with gradient accumulation of 4 (effective batch size 32), bfloat16 mixed precision, gradient checkpointing, linear learning-rate warmup over 500 steps, weight decay 0.001, and learning rate 5×10^{-5}.

4.5. CAFT loss formulation

The CAFT objective augmented standard cross-entropy classification with a continuity regularization term:

$$\mathcal{L} = \mathcal{L}_{\text{cls}} + \lambda \cdot \mathcal{L}_{\text{cont}}, \tag{1}$$

where $\mathcal{L}_{\text{cls}}$ is the cross-entropy loss over the four epithelial cell-state classes and $\mathcal{L}_{\text{cont}}$ penalizes deviation from expected adjacency structure. For each training cell x_i with assigned differentiation layer $\ell_i \in \{0, 1, 2, 3, 4\}$, the continuity loss was

$$\mathcal{L}_{\text{cont}} = -\frac{1}{N} \sum_{i=1}^{N} \sum_{\ell' \in \mathcal{A}(\ell_i)} \cos\Big(\mathbf{z}_i,\ \text{sg}(\mathbf{c}_{\ell'})\Big), \tag{2}$$

where $\mathbf{z}_i$ is the ℓ_2-normalized [CLS] embedding of cell x_i, $\mathcal{A}(\ell_i)$ denotes the adjacent layers ($\ell_i \pm 1$, clipped to $[0, 4]$), $\mathbf{c}_{\ell'}$ is the centroid of training embeddings assigned to layer ℓ', and $\text{sg}(\cdot)$ is the stop-gradient operator. Layer centroids were precomputed at the start of each epoch from detached embeddings and held fixed throughout that epoch. NON_LAYER cells ($\ell = -1$) were excluded from the continuity term but contributed to $\mathcal{L}_{\text{cls}}$.

4.6. Ordinal head formulation

To test whether the collapse of the differentiation gradient is caused by treating the four ordered layers as unordered softmax classes, Arms 5–6 replaced the softmax classification head with a cumulative ordinal head following the CORN formulation [31]. For $K = 4$ ordinal classes, the head emits $K - 1 = 3$ binary threshold logits, where threshold k predicts whether the cell has reached ordinal stage k or beyond ($P(y \geq k)$); the loss is the sum of the three binary cross-entropies, masked to epithelial cells:

$$\mathcal{L}_{\text{ord}} = \sum_{k=1}^{K-1} \text{BCE}\Big(\sigma(f_k(\mathbf{z})),\ [y \geq k]\Big), \tag{3}$$

where $f_k(\mathbf{z})$ is the k-th threshold logit on the [CLS] representation $\mathbf{z}$, σ is the sigmoid, and $[y \geq k]$ is the binary target indicating that the cell's ordinal layer is at least k. At inference the predicted class is the number of thresholds with $\sigma(f_k(\mathbf{z})) \geq 0.5$. The CORN head makes no equidistance assumption on the thresholds and yields class probabilities directly for calibration ($P(y = k) = p_k - p_{k+1}$, with $p_0 = 1$, $p_K = 0$). The ordinal-CAFT arm (Arm 6) added the CAFT continuity term ($\mathcal{L} = \mathcal{L}_{\text{ord}} + \lambda \cdot \mathcal{L}_{\text{cont}}$), with $\mathcal{L}_{\text{cont}}$ wired through a registered `TrainerCallback` so that the continuity objective was genuinely executed at each epoch start—in contrast to the archived CAFT arm, where the equivalent hook was never dispatched.

4.7. Evaluation metrics

All metrics were computed on the locked external test set using the Tier-B evaluation pipeline (corneal_fm/evaluation.py).

Primary endpoint. Donor-level macro-averaged F1 (macro-F1) on the four-class external test set, with 95% confidence intervals estimated by nonparametric bootstrap resampling of donors (n_boot = 1,000; seed = 42). Per-class precision, recall, F1, and support were reported for each arm and seed.

Secondary endpoints. (i) Differentiation-gradient rank correlation (Spearman ρ): an ordinal-structure metric computed solely from the layer-centroid geometry of the learned embeddings, without marker-gene expression. For each of the four layer classes (NON_LAYER excluded), the centroid of the cell embeddings was computed; the principal differentiation axis was defined as the first singular vector of the centered layer centroids, oriented so that pseudotime increases from LSC to differentiated; each cell was assigned a pseudotime equal to the projection of its embedding onto this axis; and ρ was the Spearman correlation between per-cell pseudotime and the a priori layer ordinal. (ii) Layer Adjacency Accuracy (LAA): the fraction of nearest-neighbor pairs in the learned embedding space belonging to adjacent or identical differentiation layers. (iii) Expected Calibration Error (ECE): the weighted mean absolute difference between predicted confidence and observed accuracy across probability bins. (iv) Internal silhouette coefficient and gap statistic were computed as diagnostic-only measures and were not used for model selection or promotion decisions.

Layer-resolved gradient measurement. At inference, all thirteen hidden states (embedding output L0 and layers L1–L12) were collected for each of the 14,775 external-test epithelial cells, and the differentiation-gradient ρ was recomputed at each depth from the position-0 [CLS] representation (scripts/measure_layerwise_rho.py). A sanity gate required the L12 ρ of every model to reproduce the corresponding single-value measurement within 10^{-6} (all seven models passed bit-exactly). At L0 the position-0 [CLS] representation is constant across cells, so ρ is undefined (NaN) and was excluded. Cell-level bootstrap 95% confidence intervals at L12 were computed with n_boot = 1,000 resamples (seed = 42; scripts/bootstrap_layerwise_rho.py).

Trajectory inference. Diffusion pseudotime (scanpy DPT) was computed on PCA-reduced external embeddings (n_pcs = 30, n_neighbors = 30, n_dcs = 10), rooted at the LSC class, and correlated (Spearman) with the reference layer-ordering pseudotime for each arm (scripts/analyze_trajectory_continuity.py).

TAC sensitivity analysis. Because the TAC class is extremely rare (171/14,775 cells; 1.2%), donor-level macro-F1 was recomputed excluding TAC (three classes), with the

same bootstrap procedure, to assess whether the rare class drives the primary endpoint (scripts/analyze_tac_sensitivity.py).

4.8. CAFT promotion criteria

CAFT (Arm 4) was promoted over naive fine-tuning (Arm 3) only if all four criteria were satisfied simultaneously: (1) external continuity improvement (ρ or LAA) relative to Arm 3; (2) non-inferior classification (bootstrap 95% CI overlap or non-inferiority within a prespecified margin); (3) seed and donor consistency of the improvement direction; and (4) survival of the locked provenance audit and external-source disjointness verification. Failure of any single criterion retained Arm 3 as the preferred strategy, with CAFT reported as an unsuccessful candidate; no post hoc tuning or selective reporting was permitted.

4.9. Provenance and contamination audit

All computational artifacts were version-locked prior to analysis: model weights (retrieved 2026-07-23), training and test data files (downloaded 2026-07-22), and SHA-256 checksums for model weights and train/test AnnData were recorded, and the fair-split manifest was generated before any training. Because Geneformer was pretrained on Genecorpus-104M, we assessed whether corneal or ocular surface datasets from our training atlas might have been included in the pretraining corpus. For Genecorpus-30M (V1; 561 datasets with a public manifest), all 43 eye-related datasets were identified; all were retinal or retinal pigment epithelium datasets, with zero corneal datasets, and all eight GEO accessions corresponding to our training studies returned no matches. For Genecorpus-104M (V2; 2,903 datasets), no public machine-readable dataset manifest was available and the V2 publication [9] was behind a paywall, precluding systematic cross-referencing; the V2 audit was therefore deemed not executable. This limitation does not affect the validity of the external test evaluation, as GSE153515 was confirmed source-disjoint from the training atlas independently of the pretraining corpus.

CAFT implementation audit. A post hoc audit of the six archived fine-tuning runs (the archived copies of scripts/caft_gf.py in the six save directories, byte-identical to the current script; MD5 8241ff23dea92ff1789e23c569108f2a) established that the continuity term $\mathcal{L}_{\text{cont}}$ never executed in any run. In Transformers 4.57, the `on_epoch_begin` hook is dispatched only to registered `TrainerCallback` objects; `CAFTTrainer.on_epoch_begin` is a plain method that no callback ever invokes, so `layer_centroids` was never populated and `compute_loss` always fell back to $\mathcal{L}_{\text{cls}}$. The training history of every run contains only the plain `loss` key, with no `L_cont` or `caft_loss` entries. The $\lambda = 0.5$ checkpoints therefore trained pure cross-entropy, identical in objective to the $\lambda = 0$ arm, and the separation between the two groups' final-layer ρ (naive mean -0.008; $\lambda = 0.5$ mean $+0.210$) is attributable to run-to-run nondeterminism, bfloat16 training over 7,473 steps and stochastic data preparation that permutes class identifiers and row order between runs, rather than to the CAFT objective. The locked results_*.json historical values are retained unchanged; ECE values recomputed with the corrected half-open binning convention are recorded in artifacts/paper1_pipeline_audit/ece_corrected.json. A correctly wired continuity objective was subsequently tested in the ordinal-CAFT arm (Section 2.5, Arm 6; Results, ordinal head), where it executed genuinely but conferred no measurable gradient benefit beyond the ordinal head itself; the layer-aware HPA-FT arms remain pending.

4.10. Statistical analysis plan

All statistical analyses were prespecified in the locked evaluation configuration prior to model training. The primary comparison was between Arm 3 and Arm 4, with Arms 1 and 2 as reference baselines; because the design involved two planned pairwise comparisons of primary interest (Arm 2 vs. Arm 1; Arm 4 vs. Arm 3) rather than an all-pairs comparison, no multiplicity correction was applied. Uncertainty was quantified by nonparametric bootstrap resampling at the donor level (n_boot = 1,000; seed = 42). Model stability was assessed across three independent training seeds, reported as mean ± range. All hypothesis tests were two-sided; $p < 0.05$ was considered significant. No interim analyses or adaptive stopping rules were employed.

4.11. Software and reproducibility

All analyses were performed in Python 3.10, using Scanpy [47], harmonypy [46, 48], Hugging Face Transformers [49], PyTorch [50], and scikit-learn [51]. The CAFT training pipeline was implemented as a custom extension of the Hugging Face Trainer class (corneal_fm/caft.py); evaluation used the Tier-B module (corneal_fm/evaluation.py); external annotation used the locked pipeline script and configuration file.

Resource availability

Lead contact. Further information and requests for resources should be directed to and will be fulfilled by the lead contact, Qi Dai (dq@mail.eye.ac.cn).

Materials availability. This study did not generate new unique reagents or biological materials; all data are public (below).

Data and code availability.

- **Training data.** CELLxGENE Discover collection 0f7d022a-46c7-4e64-be4c-e34adbb78089 (Single cell atlas of the human ocular surface; Human Ocular Surface Atlas), current dataset-version of dataset 8ef06ea4-8d0f-4303-a652-7d4604402b1f, distributed as file 30127f02-d0b9-4b56-bf4c-e5b77e4989bd.h5ad; MD5 39f098367af95bc3e1ee92528ceb251e.
- **External test data.** GEO GSE153515 [3]; 17,587 cells from two donors (GSM4646295, GSM4646296).
- **Model.** Geneformer-V2-104M, Hugging Face model hub (ctheodoris/Geneformer; snapshot commit 04c2b2e84da7c0f385c3f9ad8f3ec24bab6650e5).
- **Code.** Analysis scripts are available in a public GitHub repository [to be created upon acceptance].
- **Evaluation scripts.** scripts/run_tier_b_eval.py; scripts/run_tier_b_baselines.py; scripts/measure_layerwise_rho.py; scripts/bootstrap_layerwise_rho.py; scripts/analyze_trajectory_continuity.py; scripts/analyze_tac_sensitivity.py (listed individually below for compatibility).

- **Results.** artifacts/paper1_tier_b_eval/*.json (six fine-tuned-arm and two baseline-arm result files).
- **Reproducibility.** All experiments are reproducible with fixed random seeds (42, 123, 456), locked configuration files (config/external_test_gse153515.yml), and the SHA-256-locked fair-split manifest.

References

[1] G. Cotsarelis, S.-Z. Cheng, G. Dong, T.-T. Sun, R. M. Lavker, Existence of slow-cycling limbal epithelial basal cells that can be preferentially stimulated to proliferate: implications on epithelial stem cells, Cell 57 (2) (1989) 201–209. doi:10.1016/0092-8674(89)90958-6.

[2] H. S. Dua, A. Azuara-Blanco, Limbal stem cells of the corneal epithelium, Survey of Ophthalmology 44 (5) (2000) 415–425. doi:10.1016/S0039-6257(00)00109-0.

[3] D.-Q. Li, S. Kim, J.-M. Li, Q. Gao, J. Choi, F. Bian, J. Hu, Y. Zhang, R. Lu, Y. Lu, S. C. Pflugfelder, Single-cell transcriptomics identifies limbal stem cell population and cell types mapping its differentiation trajectory in limbal basal epithelium of human cornea, The Ocular Surface 20 (2021) 20–32. doi:10.1016/j.jtos.2020.12.004.

[4] J.-M. Li, S. Kim, D.-Q. Li, Q. Gao, J. Choi, F. Bian, J. Hu, Y. Zhang, R. Lu, S. C. Pflugfelder, Single-cell transcriptomics identifies a unique entity and signature markers of transit-amplifying cells in human corneal limbus, Investigative Ophthalmology & Visual Science 62 (9) (2021) 36. doi:10.1167/iovs.62.9.36.

[5] G. Maiti, M. Alameen, D. Nayak, M. S. Sridhar, S. Ghosh, Single cell rna-seq of human cornea organoids identifies cell fates of a developing immature cornea, PNAS Nexus 1 (5) (2022) pgac246. doi:10.1093/pnasnexus/pgac246.

[6] R. Chen, CZI Cell Science / Human Cell Atlas Eye Network, Single-cell atlas of the human ocular surface, CELLxGENE Discover collection 0f7d022a-46c7-4e64-be4c-e34adbb78089, accessed 2026. Pre-publication consortium atlas; cited as a collection (2026).
URL https://cellxgene.cziscience.com/collections/0f7d022a-46c7-4e64-be4c-e34adbb78089

[7] J. A. Arts, S. Fallo, M. S. Florencio, J. G. A. Smits, D. Lima Cunha, J. A. Y. Roubroeks, et al., Prediction of cell states and key transcription factors of the human cornea through integrated single-cell omics analyses, PNAS Nexus 4 (8) (2025) pgaf235. doi:10.1093/pnasnexus/pgaf235.

[8] C. V. Theodoris, L. Xiao, A. Chopra, M. D. Chaffin, Z. R. Al Sayed, M. C. Hill, H. Mantineo, E. M. Brydon, Z. Zeng, X. S. Liu, P. T. Ellinor, Transfer learning enables predictions in network biology, Nature 618 (7965) (2023) 616–624. doi:10.1038/s41586-023-06139-9.

[9] H. Chen, et al., Scaling and quantization of large-scale foundation model enables resource-efficient predictions in network biology, Nature Computational Science 6 (5) (2026) 450–463. doi:10.1038/s43588-026-00972-4.

[10] H. Cui, C. Wang, H. Maan, K. Pang, F. Luo, N. Duan, B. Wang, scgpt: toward building a foundation model for single-cell multi-omics using generative ai, Nature Methods 21 (8) (2024) 1470–1480. doi:10.1038/s41592-024-02201-0.

[11] F. Yang, W. Wang, F. Wang, Y. Fang, D. Tang, J. Huang, H. Lu, J. Yao, scbert as a large-scale pretrained deep language model for cell type annotation of single-cell rna-seq data, Nature Machine Intelligence 4 (10) (2022) 852–866. doi:10.1038/s42256-022-00534-z.

[12] S. Hou, P. Yang, W. Ma, J.-X. Wang, X. Zhou, A unified framework enables accessible deployment and comprehensive benchmarking of single-cell foundation models, bioRxiv preprint (2026). doi:10.64898/2026.01.06.698060.

[13] J. Wu, Q. Ye, Y. Wang, et al., Biology-driven insights into the power of single-cell foundation models, Genome Biology 26 (2025) 334. doi:10.1186/s13059-025-03781-6.

[14] T. Liu, K. Li, Y. Wang, H. Li, H. Zhao, et al., Evaluating the utilities of foundation models in single-cell data analysis, Advanced Science 13 (2026) e14490. doi:10.1002/advs.202514490.

[15] P. Qiu, Q. Chen, H. Qin, S. Fang, Y. Zhang, Y. Zhang, et al., Biollm: A standardized framework for integrating and benchmarking single-cell foundation models, Patterns 6 (8) (2025) 101326. doi:10.1016/j.patter.2025.101326.

[16] C. Wang, Y. T. Chen, et al., Evaluating large language models for evidence-based clinical question answering, Patterns 7 (5) (2026) 101519. doi:10.1016/j.patter.2026.101519.

[17] Z. Ma, J. Li, J. Hu, B. Yu, Uncovering smooth structures in single-cell data with neighbor embeddings guided by predictability-computability-stability, Patterns 7 (7) (2026) 101621. doi:10.1016/j.patter.2026.101621.

[18] F. He, R. Fei, J. E. Krull, Y. Yu, X. Zhang, X. Wang, et al., Harnessing the power of single-cell large language models with parameter-efficient fine-tuning using scpeft, Nature Machine Intelligence 8 (2026) 118–133. doi:10.1038/s42256-025-01170-z.

[19] X. Zhou, Z. Wang, Y. Ling, Q. Tian, Z. Zhang, Y. Li, L. Chen, P. Zhou, Benchmarking zero-shot single-cell foundation model embeddings for cellular dynamics reconstruction, bioRxiv preprint (2026). doi:10.64898/2026.03.10.710748.

[20] S. Pan, E. Withnell, C. Celik, M. Secrier, From cell types to cell states: adapting foundation models for cellular plasticity, bioRxiv preprint, v2 January 2026 (2024). doi:10.1101/2024.08.16.608311.

[21] I. Kendiukhov, et al., Attention layers in single-cell foundation models organize as protein–protein interactions in early layers and transcriptional regulation in late layers, BMC Genomics 27 (2026) 12965. doi:10.1186/s12864-026-12965-8.

[22] I. Kendiukhov, et al., Residual-stream geometry in single-cell foundation models, BMC Bioinformatics 27 (2026) 6538. doi:10.1186/s12859-026-06538-5.

[23] A. Kazantsev, et al., Embeddings from language models are good learners for single-cell data analysis, Patterns 7 (2) (2026) 101431. doi:10.1016/j.patter.2025.101431.

[24] M. Teles, et al., Unraveling learning characteristics of transformer models for molecular design, Patterns 6 (12) (2025) 101392. doi:10.1016/j.patter.2025.101392.

[25] W. A. Van Amsterdam, et al., When accurate prediction models yield harmful self-fulfilling prophecies, Patterns 6 (4) (2025) 101229. doi:10.1016/j.patter.2025.101229.

[26] J. Burridge, et al., The inadequacy of offline large language model evaluations, Patterns 6 (12) (2025) 101397. doi:10.1016/j.patter.2025.101397.

[27] C. Ma, T. Obuchi, T. Tanaka, Neural collapse in cumulative link models for ordinal regression: An analysis with unconstrained feature model, arXiv preprint, neurIPS 2025 (camera-ready); arXiv:2506.05801 (2025).

[28] V. Y. Civale, R. Semeraro, A. D. Bagdanov, A. Magi, Intermediate layers encode optimal biological representations in single-cell foundation models, arXiv preprint, arXiv:2604.14838 (2026).

[29] K. Inecik, A. Rose, M. Haniffa, M. Luecken, F. J. Theis, Beyond visual inspection: Principled benchmarking of single-cell trajectory representations with sctram, bioRxiv preprint (2025). doi:10.1101/2025.06.23.661141.

[30] K. Z. Kedzierska, L. Crawford, A. P. Amini, A. X. Lu, Zero-shot evaluation reveals limitations of single-cell foundation models, Genome Biology 26 (2025) 101. doi:10.1186/s13059-025-03574-x.

[31] W. Cao, V. Mirjalili, S. Raschka, Rank consistent ordinal regression for neural networks with application to age estimation, Pattern Recognition Letters 140 (2020) 325–331. doi:10.1016/j.patrec.2020.11.008.

[32] E.-I. Son, J.-T. Kim, S.-W. Seo, Ordinal neural collapse as a representation prior for visual navigation, arXiv preprint, arXiv:2606.26839 (2026).

[33] A. Aghajanyan, A. Shrivastava, A. Gupta, N. Goyal, L. Zettlemoyer, S. Gupta, Better fine-tuning by reducing representational collapse, arXiv preprint, iCLR 2021; arXiv:2008.03156 (2021).

[34] V. Papyan, X. Y. Han, D. L. Donoho, Prevalence of neural collapse during the terminal phase of deep learning training, Proceedings of the National Academy of Sciences 117 (40) (2020) 24652–24663. doi:10.1073/pnas.2015509117.

[35] G. Andriopoulos, Z. Dong, L. Guo, Z. Zhao, K. Ross, The prevalence of neural collapse in neural multivariate regression, in: Advances in Neural Information Processing Systems 37 (NeurIPS 2024), Vol. 37, 2024, pp. 126417–126451.

[36] A. Kumar, A. Raghunathan, R. Jones, T. Ma, P. Liang, Fine-tuning can distort pretrained features and underperform out-of-distribution, arXiv preprint arXiv:2202.10054, published at ICLR 2022 (2022).

[37] A. Merchant, D. Radev, M. Gastaldi, What happens to BERT embeddings during fine-tuning?, in: Proceedings of the Third BlackboxNLP Workshop on Analyzing and Interpreting Neural Networks for NLP, Association for Computational Linguistics, Online, 2020, pp. 33–44. doi:10.18653/v1/2020.blackboxnlp-1.3.

[38] Y. Lee, A. S. Chen, F. Tajwar, A. Kumar, H. Yao, P. Liang, C. Finn, Surgical fine-tuning improves adaptation to distribution shifts, arXiv preprint, iCLR 2023; arXiv:2210.11466 (2023).

[39] I. Kendiukhov, Multi-dimensional spectral geometry of biological knowledge in single-cell transformer representations, arXiv preprint, arXiv:2602.22247 (2026).

[40] P. Vig, R.-C. Wu, W. Lotter, Do pathology foundation models encode disease progression? a pseudotime analysis of visual representations, arXiv preprint, arXiv:2601.21334 (2026).

[41] E. Pareja-Lorente, P. Aloy, Task-adapted biological foundation models uncover perturbation-centric representations, bioRxiv preprint (2026). doi:10.64898/2026.06.30.735584.

[42] J. Collin, R. Queen, D. Zerti, B. Dorgau, M. Georgiou, I. Djidrovski, R. Hussain, J. M. Coxhead, A. Joseph, P. Rooney, S. Lisgo, F. Figueiredo, L. Armstrong, M. Lako, A single cell atlas of human cornea that defines its development, limbal progenitor cells and their interactions with the immune cells, The Ocular Surface 21 (2021) 279–298. doi:10.1016/j.jtos.2021.03.010.

[43] S. Dou, Q. Wang, X. Qi, B. Zhang, H. Jiang, S. Chen, J. Li, M. Li, Molecular identity of human limbal heterogeneity involved in corneal homeostasis and privilege, The Ocular Surface 21 (2021) 206–220. doi:10.1016/j.jtos.2021.04.010.

[44] P. Català, N. Groen, J. A. Dehnen, J. S. Soares, A. J. H. van Velthoven, R. M. M. A. Nuijts, M. M. Dickman, V. L. S. LaPointe, Single cell transcriptomics reveals the heterogeneity of the human cornea to identify novel markers of the limbus and stroma, Scientific Reports 11 (1) (2021) 21727. doi:10.1038/s41598-021-01015-w.

[45] P. Gautam, K. Hamashima, Y. Chen, Y. Zeng, B. Makovoz, B. H. Parikh, Y. Lee, K. A. Lau, B. Lim, B. X. Ng, et al., Multi-species single-cell transcriptomic analysis of ocular compartment regulons, Nature Communications 12 (1) (2021) 5675. doi:10.1038/s41467-021-25968-8.

[46] I. Korsunsky, N. Millard, J. Fan, K. Slowikowski, F. Zhang, K. Wei, Y. Baglaenko, M. Brenner, P.-R. Loh, S. Raychaudhuri, Fast, sensitive and accurate integration of single-cell data with harmony, Nature Methods 16 (12) (2019) 1289–1296. doi:10.1038/s41592-019-0619-0.

[47] F. A. Wolf, P. Angerer, F. J. Theis, Scanpy: large-scale single-cell gene expression data analysis, Genome Biology 19 (1) (2018) 15. doi:10.1186/s13059-017-1382-0.

[48] K. Slowikowski, Korsunsky, Ilya, harmonypy: Python implementation of harmony, GitHub repository, accessed 2026 (2020).
URL https://github.com/slowkow/harmonypy

[49] T. Wolf, L. Debut, V. Sanh, J. Chaumond, C. Delangue, A. Moi, P. Cistac, T. Rault, R. Louf, M. Funtowicz, J. Davison, S. Shleifer, P. von Platen, C. Ma, Y. Jernite, J. Plu, C. Xu, T. Le Scao, S. Gugger, M. Drame, Q. Lhoest, A. M. Rush, Huggingface's transformers: State-of-the-art natural language processing, in: Proceedings of the 2020 Conference on Empirical Methods in Natural Language Processing: System Demonstrations, 2020, pp. 38–45. doi:10.18653/v1/2020.emnlp-demos.6.

[50] A. Paszke, S. Gross, F. Massa, A. Lerer, J. Bradbury, G. Chanan, T. Killeen, Z. Lin, N. Gimelshein, L. Antiga, A. Desmaison, A. Kopf, E. Yang, Z. DeVito, M. Raison, A. Tejani, S. Chilamkurthy, B. Steiner, L. Fang, J. Bai, S. Chintala, Pytorch: An imperative style, high-performance deep learning library, in: Advances in Neural Information Processing Systems 32, 2019, pp. 8026–8037.

[51] F. Pedregosa, G. Varoquaux, A. Gramfort, V. Michel, B. Thirion, O. Grisel, M. Blondel, P. Prettenhofer, R. Weiss, V. Dubourg, J. Vanderplas, A. Passos, D. Cournapeau, M. Brucher, M. Perrot, É. Duchesnay, Scikit-learn: Machine learning in python, Journal of Machine Learning Research 12 (2011) 2825–2830.
URL http://jmlr.org/papers/v12/pedregosa11a.html